

Lecture 7

Combinatorics; More Combinatorics Examples; Bayes' Theorem and Independence

DSC 40A, Summer 2026

Lecture 7 Part 1

Combinatorics

DSC 40A, Summer 2026

Agenda

- Combinatorics.
- Start some examples.

Remember, we've posted **many** probability resources on the resources tab of the course website. These will come in handy! Specific resources you should look at:

- The DSC 40A probability roadmap, written by Janine Tiefenbruck.
- The textbook [Theory Meets Data](#), which explains many of the same ideas and contains more practice problems.

For today's lectures specifically, there are two supplementary videos I created that you should watch. Both are linked in [this playlist](#).

Question 🤔

Answer in chat!

Remember, you can always ask questions in Q&A!

Combinatorics

Motivation

- Many problems in probability involve counting.
 - Suppose I flip a fair coin 100 times. What's the probability I see 34 heads?
 - Suppose I draw 3 cards from a 52 card deck. What's the probability they all are all from the same suit?
- In order to solve such problems, we first need to learn how to count.
- The area of math that deals with counting is called **combinatorics**.

Selecting elements (i.e. sampling)

- Many experiments involve choosing k elements randomly from a group of n possible elements. This group is called a **population**.
 - If drawing cards from a deck, the population is the deck of all cards.
 - If selecting people from DSC 40A, the population is everyone in DSC 40A.
- Two decisions:
 - Do we select elements with or without **replacement**?
 - Does the **order** in which things are selected matter?

Sequences

- A **sequence** of length k is obtained by selecting k elements from a group of n possible elements **with replacement (i.e. repetition is allowed)**, such that **order matters**.
- **Example:** Draw a card (from a standard 52-card deck), put it back in the deck, and repeat 4 times. How many such sequences are there?
- **Example:** A UCSD PID starts with "A" then has 8 digits. How many UCSD PIDs are possible?

Sequences

In general, the number of ways to select k elements from a group of n possible elements **with replacement (i.e. repetition is allowed)** and **order matters** is n^k .

Permutations

- A **permutation** is obtained by selecting k elements from a group of n possible elements **without replacement (i.e. repetition is not allowed)**, such that **order matters**.
- **Example:** Draw 4 cards, without replacement, from a standard 52-card deck. How many such permutations are there?
- **Example:** How many ways are there to select a president, vice president, and secretary from a group of 8 people?

Permutations

- In general, the number of ways to select k elements from a group of n possible elements **without replacement (i.e. repetition is not allowed)** and **order matters** is:

$$P(n, k) = (n)(n - 1) \dots (n - k + 1)$$

- To simplify: recall that the definition of $n!$ is:
- Given this, we can write:

Special case of permutations

- Suppose we have n people. The total number of ways I can rearrange these n people in a line is:

- This is consistent with the formula:

$$P(n, n) = \frac{n!}{(n - n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$$

- **Followup:** How many ways are there to arrange n people in a circle?

Combinations

- A **combination** is a set of k elements selected from a group of n possible elements **without replacement (i.e. repetition is not allowed)**, such that **order does not matter**.
- **Example:** There are 4 ice cream flavors. In how many ways can you pick two different flavors?

From permutations to combinations

- There is a close connection between:
 - the number of **permutations** of k elements selected from a group of n , and
 - the number of **combinations** of k elements selected from a group of n .

$$\# \text{ combinations} = \frac{\# \text{ permutations}}{\# \text{ orderings of } k \text{ elements}}$$

- Since $\# \text{ permutations} = \frac{n!}{(n-k)!}$ and $\# \text{ orderings of } k \text{ elements} = k!$, we have:

$$C(n, k) = \binom{n}{k} = \frac{n!}{(n-k)!k!}$$

Combinations

In general, the number of ways to select k elements from a group of n elements **without replacement (i.e. repetition is not allowed) and **order does not matter** is:

$$\boxed{\binom{n}{k} = \frac{n!}{(n-k)!k!}}$$

The symbol $\binom{n}{k}$ is pronounced " n choose k ", and is also known as the **binomial coefficient**.

Example: Committees

- How many ways are there to select a president, vice president, and secretary from a group of 8 people?
- How many ways are there to select a committee of 3 people from a group of 8 people?
- If you're ever confused about the difference between permutations and combinations, **come back to this example.**
- More generally, don't jump straight to a formula: think about what the question is asking for.

Aside: Simplifying $\binom{n}{k}$

It's true that:

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}$$

However, when asked to simplify the value of $\binom{n}{k}$, do so strategically!

Question 🤔

Answer in chat

A domino consists of two faces, each with anywhere between 0 and 6 dots. A set of dominoes consists of every possible combination of dots on each face.

How many dominoes are in a set of dominoes?

- A. $\binom{7}{2}$
- B. $\binom{7}{1} + \binom{7}{2}$
- C. $P(7, 2)$
- D. $\frac{P(7,2)}{P(7,1)} 7!$

Summary

Suppose we want to select k elements from a group of n possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

	Yes, order matters	No, order doesn't matter
With replacement Repetition allowed	n^k possible sequences	more complicated: watch this video *
Without replacement Repetition not allowed	$\frac{n!}{(n-k)!}$ possible permutations	$\binom{n}{k}$ combinations

**or see the previous slide.*

Combinatorics as a tool for probability

- If S is a sample space consisting of equally-likely outcomes, and A is an event, then $\mathbb{P}(A) = \frac{|A|}{|S|}$.
- In many examples, this will boil down to using permutations and/or combinations to count $|A|$ and $|S|$.
- **Tip:** Before starting a probability problem, always think about what the sample space S is!

More examples

Watch the video linked below!

The following few slides **will not** be covered in lecture (unless we have extra time). Instead, they're covered in [this walkthrough video](#).

Example: Pets

Part 1: We have 12 pets: 5 dogs and 7 cats. In how many ways can we select 4 pets?

Example: Pets

Part 2: We have 12 pets: 5 dogs and 7 cats. In how many ways can we select 4 pets such that we have...

1. 2 dogs and 2 cats?
2. 3 dogs and 1 cat?
3. At least 2 dogs?

Example: Pets

Part 3: We have 12 pets: 5 dogs and 7 cats. We randomly select 4 pets. What's the probability that we selected at least 2 dogs?

Example: A fair coin

Suppose we flip a fair coin 10 times.

1. What is the probability that we see the specific sequence THTTHTHHTH?
2. What is the probability that we see an equal number of heads and tails?

Example: An unfair coin

Suppose we flip an **unfair** coin 10 times. The coin is biased such that for each flip, $\mathbb{P}(\text{heads}) = \frac{1}{3}$.

1. What is the probability that we see the specific sequence THTTHTHHTH?
2. What is the probability that we see an equal number of heads and tails?

Announcements

Announcements

- Homework 6 is due on **Friday at 11:59PM**.
 - Combinatorics is hard! Come to office hours.
 - No seriously, come to office hours.
 - Like actually. Use them. They are FOR YOU.
 - If none of the available times work for you, make a private piazza post.
- Groupwork 3 will be released after lecture today.
 - Has your group signed up for a check off meeting this week?
- Final is coming up
 - Proctoring guide is released. Please make sure to read it in full.
 - The formatting guide will come out next week when we are able to give you a better accurate idea of the max number of pages.

Lecture 7 Part 2

More Combinatorics Examples

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Agenda

- Example: Selecting students.
- Example: An unfair coin.
- Example: Poker.
- Example: Chess.

Example: Selecting students

Summary

Suppose we want to select k elements from a group of n possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

	Yes, order matters	No, order doesn't matter
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Without replacement Repetition not allowed	$\frac{n!}{(n-k)!}$ possible permutations	$\binom{n}{k}$ combinations

Using the binomial coefficient, $\binom{n}{k}$

As an example, let's evaluate:

$$\binom{9}{2} + \binom{9}{3}$$

Proof of $\binom{9}{2} + \binom{9}{3}$

Proof of $\binom{9}{2} + \binom{9}{3}$

Combinatorics as a tool for probability

- If S is a sample space consisting of equally-likely outcomes, and A is an event, then $\mathbb{P}(A) = \frac{|A|}{|S|}$.
- In many examples, this will boil down to using permutations and/or combinations to count $|A|$ and $|S|$.
- **Tip:** Before starting a probability problem, always think about what the sample space S is!

Overview: Selecting students

We're going to answer the same question using several different techniques.

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random without replacement. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 1: Using permutations

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 2: Using permutations and the complement

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 3: Using combinations

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Selecting students

Method 4: The "easy" way

There are 20 students in a class. Avi is one of them. Suppose we select 5 students in the class uniformly at random **without replacement**. What is the probability that Avi is among the 5 selected students?

Question 🤔

Answer in chat

With vs. without replacement

We've determined that a probability that a random sample of 5 students from a class of 20 **without replacement** contains Avi (one student in particular) is $\frac{1}{4}$.

Suppose we instead sampled **with replacement**. Would the resulting probability be equal to, greater than, or less than $\frac{1}{4}$?

- A. Equal to.
- B. Greater than.
- C. Less than.

Example: An unfair coin

Followup: An unfair coin

In the video you were asked to watch, we flipped an **unfair** coin 10 times, where the coin was biased such that for each flip, $\mathbb{P}(\text{heads}) = \frac{1}{3}$.

1. What is the probability that we see the specific sequence HHHHTTTTTT?
2. What is the probability that we see exactly 4 heads?

Followup: An unfair coin

3. What is the probability that we see exactly k heads, where $0 \leq k \leq 10$?

4. What is the probability that we see *at least* k heads, where $0 \leq k \leq 10$?

Example: Poker

Deck of cards

- There are **52** cards in a standard deck.
- Each card has a suit (4 possibilities) and a value (13 possibilities).

♥: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♦: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♣: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

♠: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

- In poker, each player is dealt 5 cards, called a **hand**. The order of cards in a hand does not matter.

Deck of cards

1. How many 5 card hands are there in poker?
2. How many 5 card hands are there where all cards are of the same suit (i.e. a "flush")?

Deck of cards

3. How many 5 card hands are there that include four cards of the same value (i.e. a "four-of-a-kind")?

Deck of cards

4. How many 5 card hands are there that have all card values consecutive (i.e. a "straight")?

Deck of cards

5. How many 5 card hands are there that have all card values consecutive and of the same suit (i.e. a "straight flush")?

Deck of cards

6. How many 5 card hands are there that include exactly one pair of values (e.g. aabcd)?

Example: Chess

Chess pieces

(Source: [Spring 2023 Midterm 2, Problem 3](#))

A set of chess pieces has 32 pieces. 16 of these are black and 16 of these are white. **In each color**, the 16 pieces are:

8 pawns, 2 bishops, 2 knights, 2 rooks, 1 queen, and 1 king

When there are multiple pieces of a given color and type (for example, 8 white pawns), we will assume they are **indistinguishable** from one another.

In this problem, a **lineup** is a way of arranging items in a straight line.

Chess pieces

1. A chess player lines up all 16 **white pieces** from the set of chess pieces. How many different-looking lineups can be created? Remember, some pieces look the same.

Chess pieces

2. A chess player lines up all 16 **pawns** from the set of chess pieces. How many lineups have white pawns on both ends?

Chess pieces

3. A chess player lines up all 16 **pawns** from the set of chess pieces. Assuming that each different-looking lineup is equally likely, what is the probability that the lineup has two of the same-colored pawns on both ends (both black or both white)?

Summary

Summary

Suppose we want to select k elements from a group of n possible elements. The following table summarizes whether the problem involves sequences, permutations, or combinations, along with the number of relevant orderings.

	Yes, order matters	No, order doesn't matter
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Without replacement Repetition not allowed	$\frac{n!}{(n-k)!}$ possible permutations	$\binom{n}{k}$ combinations

Lecture 7 Part 3

Bayes' Theorem and Independence

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Agenda

- Law of Total Probability.
- Bayes' Theorem.
- Independence.

Law of Total Probability

Example: Getting to school

You conduct a survey where you ask students two questions:

1. How did you get to campus today – trolley, bike, or drive? (Assume these are the only options.)
2. Were you late?

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

Question 🤔

Answer in chat.

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

What's the probability that a randomly selected person was late?

- A. 0.24 B. 0.30 C. 0.45 D. 0.50 E. None of the above.

Example: Getting to school

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

- Since everyone either takes the trolley, bikes, or drives to school, we have:

$$\mathbb{P}(\text{Late}) = \mathbb{P}(\text{Late} \cap \text{Trolley}) + \mathbb{P}(\text{Late} \cap \text{Bike}) + \mathbb{P}(\text{Late} \cap \text{Drive})$$

Question 🤔

Answer in chat.

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

Avi took the trolley to school. What is the probability that he was late?

- A. 0.06 B. 0.20 C. 0.25 D. 0.45 E. None of the above.

Example: Getting to school

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

- Since everyone either takes the trolley, bikes, or drives to school, we have:

$$\mathbb{P}(\text{Late}) = \mathbb{P}(\text{Late} \cap \text{Trolley}) + \mathbb{P}(\text{Late} \cap \text{Bike}) + \mathbb{P}(\text{Late} \cap \text{Drive})$$

- Another way of expressing the same thing:

$$\begin{aligned} \mathbb{P}(\text{Late}) = & \mathbb{P}(\text{Trolley}) \mathbb{P}(\text{Late}|\text{Trolley}) + \mathbb{P}(\text{Bike}) \mathbb{P}(\text{Late}|\text{Bike}) \\ & + \mathbb{P}(\text{Drive}) \mathbb{P}(\text{Late}|\text{Drive}) \end{aligned}$$

Partitions

- A set of events $E_1, E_2, \dots, E_k$ is a **partition** of S if:
 - $\mathbb{P}(E_i \cap E_j) = 0$ for all pairs $i \neq j$.
 - $\mathbb{P}(E_1 \cup E_2 \cup \dots \cup E_k) = 1$.
 - Equivalently, $\mathbb{P}(E_1) + \mathbb{P}(E_2) + \dots + \mathbb{P}(E_k) = 1$.
- In other words, $E_1, E_2, \dots, E_k$ is a partition of S if every outcome $s \in S$ is in **exactly** one event E_i .

Example partitions

- In getting to school, the events Trolley, Bike, and Drive.
- In getting to school, the events Late and Not Late.
- In selecting an undergraduate student at random, the events Freshman, Sophomore, Junior, and Senior, and Other.
- In rolling a die, the events Even and Odd.
- In drawing a card from a standard deck of cards, the events Spades, Clubs, Hearts, and Diamonds.
- Special case: Any event A and its complement $\bar{A}$.

The Law of Total Probability

- If A is an event and $E_1, E_2, \dots, E_k$ is a **partition** of S , then:

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap E_1) + \mathbb{P}(A \cap E_2) + \dots + \mathbb{P}(A \cap E_k) \\ &= \sum_{i=1}^k \mathbb{P}(A \cap E_i)\end{aligned}$$

The Law of Total Probability

- If A is an event and $E_1, E_2, \dots, E_k$ is a **partition** of S , then:

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap E_1) + \mathbb{P}(A \cap E_2) + \dots + \mathbb{P}(A \cap E_k) \\ &= \sum_{i=1}^k \mathbb{P}(A \cap E_i)\end{aligned}$$

- Since $\mathbb{P}(A \cap E_i) = \mathbb{P}(E_i) \cdot \mathbb{P}(A|E_i)$ by the multiplication rule, an equivalent formulation is:

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(E_1) \cdot \mathbb{P}(A|E_1) + \mathbb{P}(E_2) \cdot \mathbb{P}(A|E_2) + \dots + \mathbb{P}(E_k) \cdot \mathbb{P}(A|E_k) \\ &= \sum_{i=1}^k \mathbb{P}(E_i) \cdot \mathbb{P}(A|E_i)\end{aligned}$$

Question 🤔

Answer in chat

	Late	Not Late
Trolley	0.06	0.24
Bike	0.03	0.07
Drive	0.36	0.24

Lauren is late to school. What is the probability that she took the trolley? Choose the best answer.

- A. About 0.05 B. About 0.15 C. About 0.30 D. About 0.40

Bayes' Theorem

Example: Getting to school

- Now, suppose we don't have that entire table. Instead, all you know is:
 - $\mathbb{P}(\text{Late}) = 0.45$.
 - $\mathbb{P}(\text{Trolley}) = 0.3$.
 - $\mathbb{P}(\text{Late}|\text{Trolley}) = 0.2$.
- Can we still find $\mathbb{P}(\text{Trolley}|\text{Late})$?

Bayes' Theorem

- Recall that the multiplication rule states that:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B|A)$$

- It also states that:

$$\mathbb{P}(B \cap A) = \mathbb{P}(B) \cdot \mathbb{P}(A|B)$$

- But since $A \cap B = B \cap A$, we have that:

- Re-arranging yields **Bayes' Theorem**:

Bayes' Theorem and the Law of Total Probability

- Bayes' Theorem:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(A)}$$

- Recall from earlier, for any sample space S , B and $\bar{B}$ partition S . Using the Law of Total Probability, we can re-write $\mathbb{P}(A)$ as:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap \bar{B}) = \mathbb{P}(B) \cdot \mathbb{P}(A|B) + \mathbb{P}(\bar{B}) \cdot \mathbb{P}(A|\bar{B})$$

- This means that we can re-write Bayes' Theorem as:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B) \cdot \mathbb{P}(A|B)}{\mathbb{P}(B) \cdot \mathbb{P}(A|B) + \mathbb{P}(\bar{B}) \cdot \mathbb{P}(A|\bar{B})}$$

Example: Drug test

A manufacturer claims that its drug test will **detect steroid use**. What the company does not tell you is that 15% of all steroid-free individuals also test positive (the "false positive rate"). Suppose 10% of the Tour de France bike racers use steroids and your favorite cyclist just tested positive. What's the probability that they used steroids?

Example: Taste test

- Your friend claims to be able to correctly guess what restaurant a burger came from, after just one bite.
- The probability that she correctly identifies an In-n-Out Burger is 0.55, a Shake Shack burger is 0.75, and a Five Guys burger is 0.6.
- You buy 5 In-n-Out burgers, 4 Shake Shack burgers, and 1 Five Guys burger, choose one of the burgers randomly, and give it to her.
- **Question:** Given that she guessed it correctly, what's the probability she ate a Shake Shack burger?

Question 🤔

Answer in chat.

Consider any two events A and B . Choose the expression that's equivalent to:

$$\mathbb{P}(B|A) + \mathbb{P}(\bar{B}|A)$$

- A. $\mathbb{P}(A)$
- B. $1 - \mathbb{P}(B)$
- C. $\mathbb{P}(B)$
- D. $\mathbb{P}(\bar{B})$
- E. 1

Example: Prosecutor's fallacy

A bank was robbed yesterday by one person. Consider the following facts about the crime:

- The person who robbed the bank wore Nikes.
- Of the 10,000 other people who came to the bank yesterday, only 10 of them wore Nikes.

The prosecutor finds the prime suspect, and states that "given this evidence, the chance that the prime suspect was not at the crime scene is 1 in 1,000".

1. What is wrong with this statement?
2. Find the probability that the prime suspect is guilty given only the evidence in the exercise.

